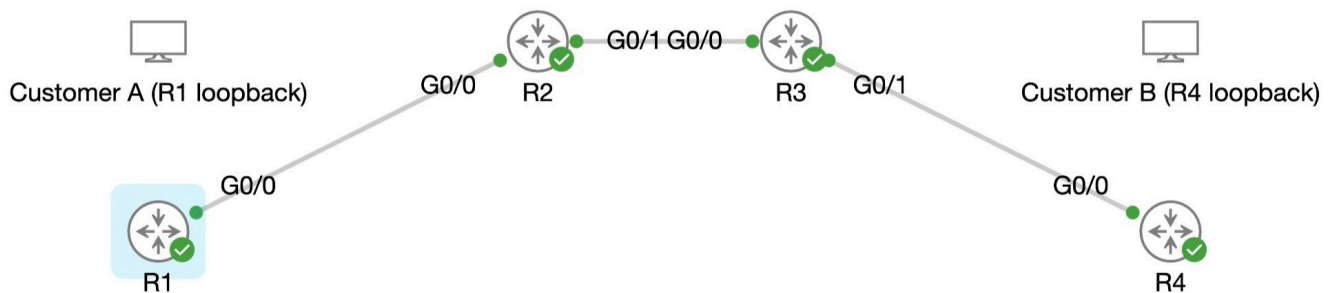


Network topology



IP Addressing Plan

R1 and R4 are provider edge routers with loopback built for simulating customer edge routers

R2 and R3 are core routers

R1 to R2 - 10.0.12.0/30

R2 to R3 - 10.0.23.0/30

R3 to R4 - 10.0.34.0/30

Routers 1,2,3,4 loopback0 - 1.1.1.1, 2.2.2.2, 3.3.3.3, 4.4.4.4

Creating loopback1 on R1 and R4 to simulate customer network :

R1 - loopback1 - 172.16.1.0/24 - this will be customer edge

R4 - loopback1 - 172.16.4.0/24 - customer edge

Two virtual router forwarding networks are created:

R1 - red

R4 - blue

Configure interfaces

- Configure interfaces

- On R1,

```
en
conf t

hostname R1

int gi0/0
 ip add 10.0.12.1 255.255.255.252
 no sh
exit

int Loopback0
 ip add 1.1.1.1 255.255.255.255
exit

int loopback1
 ip add 172.16.1.1 255.255.255.0
exit
```

- On R2,

```
en
conf t
hostname R2

int gi0/0
 ip address 10.0.12.2 255.255.255.252
 no sh
exit

int gi0/1
 ip add 10.0.23.1 255.255.255.252
 no sh
```

```
exit

int Loopback0
ip add 2.2.2.2 255.255.255.255
exit
```

- On R3,

```
en
conf t
hostname R3

int gi0/0
ip add 10.0.23.2 255.255.255.252
no sh
exit

int gi0/1
ip add 10.0.34.1 255.255.255.252
no sh
exit

int loopback0
ip add 3.3.3.3 255.255.255.255
exit
```

- On R4,

```
en
conf t
hostname R4

int gi0/0
ip add 10.0.34.2 255.255.255.252
no shutdown
exit

int loopback0
ip add 4.4.4.4 255.255.255.255
exit
```

- Configure OSPF
 - On R1,R2,R3, and R4:

```
router ospf 1
 network 0.0.0.0 255.255.255.255 area 0
exit
```

- Configure BGP
 - On R1,

```
router bgp 65000
 neighbor 4.4.4.4 remote-as 65000
 neighbor 4.4.4.4 update-source Loopback0

 address-family vpnv4
  neighbor 4.4.4.4 activate
  neighbor 4.4.4.4 send-community extended
 exit-address-family
exit
```

- On R4,

```
router bgp 65000
 bgp log-neighbor-changes
 neighbor 1.1.1.1 remote-as 65000
 neighbor 1.1.1.1 update-source Loopback0

 address-family vpnv4
  neighbor 1.1.1.1 activate
  neighbor 1.1.1.1 send-community extended
 exit-address-family
exit
```

- Configure MPLS
 - On R1,

```
int gi0/0
 mpls ip
exit
```

- On R2,

```
int gi0/0
  mpls ip
exit
```

```
int gi0/1
  mpls ip
exit
```

- On R3,

```
int gi0/0
  mpls ip
exit
```

```
int gi0/1
  mpls ip
exit
```

- On R4,

```
int gi0/0
  mpls ip
exit
```

- Configure VRF

- On R1,

```
ip vrf RED
  rd 65010:1
  route-target export 65000:100
  route-target import 65000:100
exit

router bgp 65000
  address-family ipv4 vrf RED
    redistribute connected
  exit-address-family
exit
```

- On R4,

```
ip vrf BLUE
  rd 65020:1
  route-target export 65000:200
  route-target import 65000:200
exit

router bgp 65000
  address-family ipv4 vrf BLUE
    redistribute connected
  exit-address-family
exit
```

- On R1,

```
int loopback1
  ip vrf forwarding RED
exit
```

- On R4,

```
int loopback1
  ip vrf forwarding BLUE
exit
```